



GearGuard

PPE compliance scanning,
attendance & audit reports

Flutter · flutter_bloc · YOLO-ready

Preprint + open source

Today at a glance

Live compliance ring, check-ins, violations and a 7-day trend.



GearGuard



Good night, Supervisor

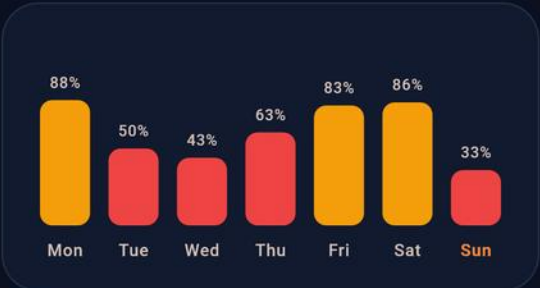
Main Site · Sunday, 20 September



2 workers have missing PPE today



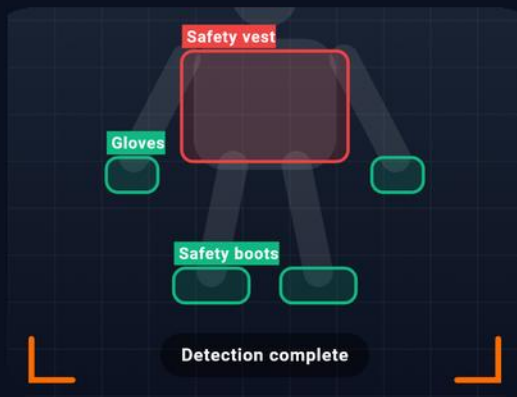
Compliance this week



- Home
- Attendance
- Scan
- Team
- Archive

Scan with confidence scores

Per-item checklist and bounding boxes. Demo detector in the app; real YOLO model tested separately.



PPE violation

Missing: Safety vest



Helmet

94%



Safety vest

31%



Gloves

88%



Safety boots

91%



Save to archive



Rescan

Discard

Scanned 22:24



Home



Attendance



Scan



Team



Archive

Attendance with PPE status

Violations first; one tap to scan whoever has not checked in.

Attendance

Sunday, 20 September

Mon
14

Tue
15

Wed
16

Thu
17

Fri
18

Sat
19

3/8
Present

1
Compliant

2
Violations

BS

Budi Santoso

In at 07:54 · Construction

Violation

AK

Ayu Kartika

In at 06:56 · Welding

Violation

SR

Siti Rahmawati

Welding

Scan

DL

Dewi Lestari

Maintenance

Scan

RP

Rizky Pratama

Construction

Scan

Home

Attendance

Scan

Team

Archive

GearGuard · PPE compliance

4/8

Team ranking & archive

Rank by compliance, filter scans, export PDF reports.

Team

8 registered employees

 Search name, ID or department

-  Everyone
-  Ranking

		Siti Rahmawati EMP-002 · Welding	100% 6 scans
		Maya Anggraini EMP-008 · Maintenance	100% 6 scans
		Budi Santoso EMP-001 · Construction	86% 7 scans
4		Dewi Lestari EMP-004 · Maintenance	67% 6 scans
5		Agus Wijaya EMP-003 · Logistics	67% 6 scans
6		Ayu Kartika EMP-006 · Welding	50%

 Add

-  Home
-  Attendance
-  Scan
-  Team
-  Archive

Verified, not just built

38

automated tests pass

75.1%

line coverage

20/20

accessibility checks (3 failed first, then fixed)

11/11

validation photos: expected PPE class found

~80 ms

per frame, YOLO TFLite on a desktop CPU

What the tests found

Honest results, including weaknesses

Occlusion

25% patch over the image → only 5/11 detected

Colour bias

plain orange patch → false "Vest" at 0.81

Threshold

letterbox keeps 11/11 up to $\tau = 0.7$; app resize drops at 0.6

Robust to

darkening · JPEG · $\pm 15^\circ$ rotation · noise · 4× downscale

Not claimed: accuracy (mAP) or usability. Both still to do.

Code & paper

GitHub

github.com/afprayogi/ppe_apk_prototype (branch main)

Paper

docs/paper: English + Bahasa Indonesia PDF

Study kit

SUS questionnaire (EN/ID) + scoring script

Feedback welcome — especially from K3 / safety officers.